WK1-PM1 – WEEK 1 PROJECT MODULE 1

LAB SETUP: VIRTUALBOX AND KALI LINUX

Project: Cybersecurity Testing Lab Environment Setup
Task ID: WK1-PM1
Student Name: Syed Raihaan
Course: Cybersecurity / Ethical Hacking
Date of Completion: 29-10-2026

1. OBJECTIVE

The objective of this task is to set up a cybersecurity testing lab environment using Oracle VirtualBox and Kali Linux. The lab is designed for authorized cybersecurity learning, penetration-testing practice, and security testing in a controlled virtual environment.

The lab configuration includes a custom NAT Network, a static IP address for Kali Linux, internet connectivity, shared clipboard, drag-and-drop functionality, a shared Downloads folder, and a virtual machine snapshot.

2. SOFTWARE AND SYSTEM REQUIREMENTS

The following software and resources were used or required for the lab setup:

Component	Details
Host Operating System	Microsoft Windows
Virtualization Software	Oracle VirtualBox
Attacker Machine	Kali Linux
Archive Utility	7-Zip
Network Type	NAT Network
Network Subnet	10.0.0.0/24
Kali Linux IP Address	10.0.0.2/24
Default Gateway	10.0.0.1
Shared Host Folder	Windows Downloads folder

Screenshot 1: *Insert a screenshot of VirtualBox Manager showing the Kali Linux virtual machine.*

3. INSTALLATION OF VIRTUALBOX

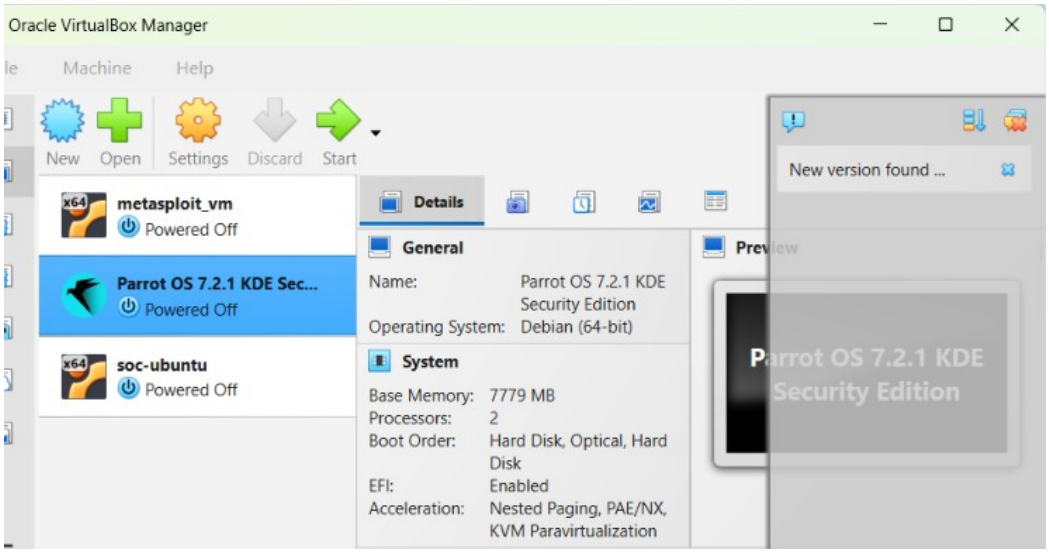
Oracle VirtualBox was used as the virtualization platform for the cybersecurity lab.

Procedure

1. Downloaded Oracle VirtualBox from the official website.
2. Installed VirtualBox on the Windows host computer.

3. Opened VirtualBox Manager and verified that the application was running.
4. Prepared the environment for importing the Kali Linux virtual machine.

Official website: https://www.virtualbox.org/wiki/Downloads



Screenshot 2: *Installed VirtualBox application (VirtualBox Manager).*

4. DOWNLOAD AND IMPORT OF KALI LINUX

Kali Linux was selected as the attacker machine for the cybersecurity testing lab.

Procedure

1. Accessed the official Kali Linux website.
2. Downloaded the prebuilt VirtualBox virtual machine image.
3. Extracted the downloaded archive using 7-Zip, if required.
4. Added the extracted virtual machine to VirtualBox Manager.
5. Configured the virtual machine's memory, processor, and network settings.
6. Started Kali Linux and verified that the operating system booted successfully.

Official website: https://www.kali.org/get-kali/#kali-virtual-machines



Screenshot 3: *Kali Linux virtual machine in VirtualBox Manager.*

Screenshot 4: *Insert a screenshot of the running Kali Linux desktop.*

5. NAT NETWORK CONFIGURATION

A custom NAT Network was configured to provide network connectivity to the virtual machines in the lab.

Network Configuration

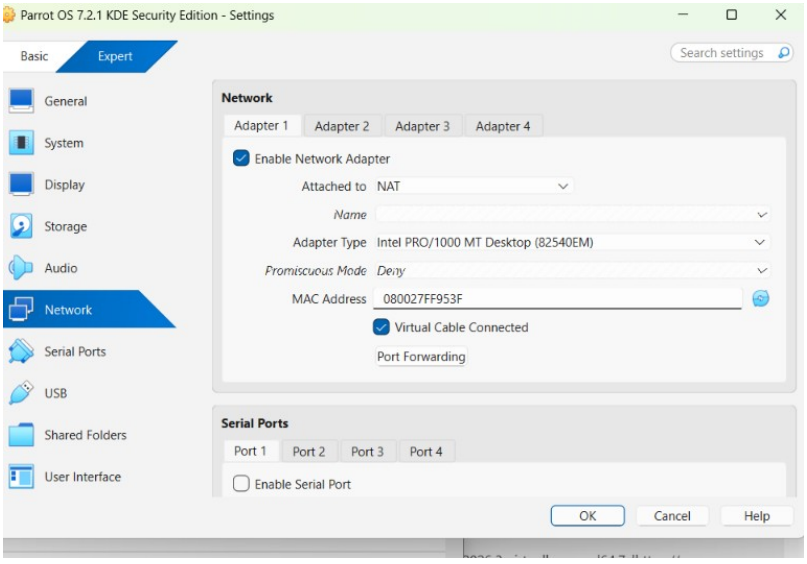
Parameter	Configuration
Network Name	CyberLab-NAT

Parameter	Configuration
Network Address	10.0.0.0
Subnet Prefix	/24
Subnet Mask	255.255.255.0
Kali Linux IP Address	10.0.0.2
Default Gateway	10.0.0.1
Network Type	NAT Network

Procedure

1. Opened VirtualBox Manager.
2. Navigated to the VirtualBox network management settings.
3. Created a custom NAT Network named CyberLab-NAT.
4. Configured the network prefix as 10.0.0.0/24.
5. Enabled the NAT Network.
6. Attached Kali Linux to the custom NAT Network through the virtual machine's network settings.

Screenshot 5: *Insert a screenshot of the NAT Network settings showing the network name and IPv4 prefix.*



Screenshot 6: *Kali Linux virtual machine settings showing Adapter 1 network attachment.*

6. KALI LINUX IP ADDRESS CONFIGURATION

The Kali Linux virtual machine was configured to use the static IPv4 address specified in the assignment.

Configuration Details

- IP address: 10.0.0.2
- Prefix length: /24

- Subnet mask: 255.255.255.0
- Default gateway: 10.0.0.1
- DNS servers: 10.0.0.1 and 8.8.8.8

Configuration Procedure

The active NetworkManager connection was identified using:

```
nmcli connection show
```

The connection was configured with the static IP address using:

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.0.0.2/24 \
  ipv4.gateway 10.0.0.1 \
  ipv4.dns "10.0.0.1 8.8.8.8"
```

The connection was then activated:

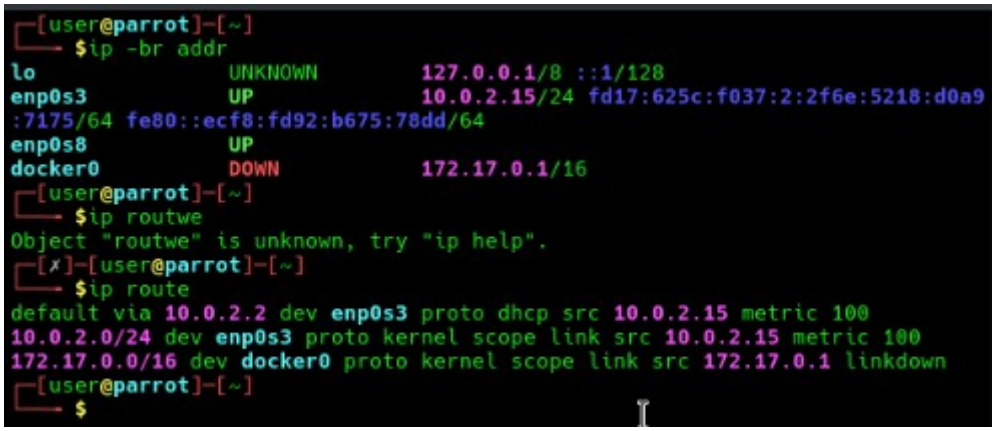
```
sudo nmcli connection up "Wired connection 1"
```

If the connection has a different name, the command must use the actual name shown by NetworkManager.

Verification

The following commands were used to inspect the network configuration:

```
ip -br addr
ip route
```



Screenshot 7: *Kali Linux IP address and network interface, with the routing table.*

Screenshot 8: *Insert a screenshot showing the default gateway in the routing table.*

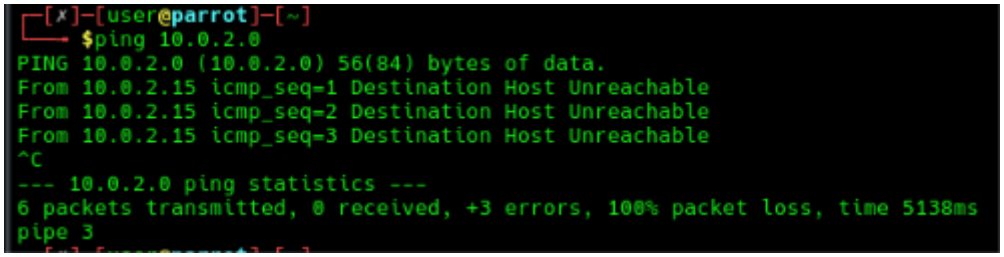
7. INTERNET CONNECTIVITY VERIFICATION

Internet connectivity was tested from the Kali Linux virtual machine.

Test 1: Gateway Connectivity

```
ping -c 4 10.0.0.1
```

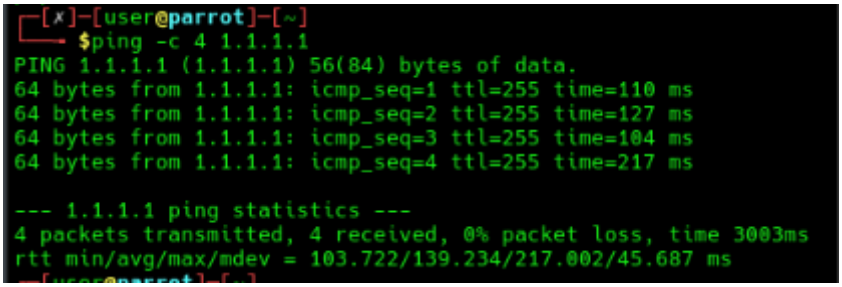
Result:



Test 2: External IP Connectivity

```
ping -c 4 1.1.1.1
```

Result:



Test 3: DNS Resolution

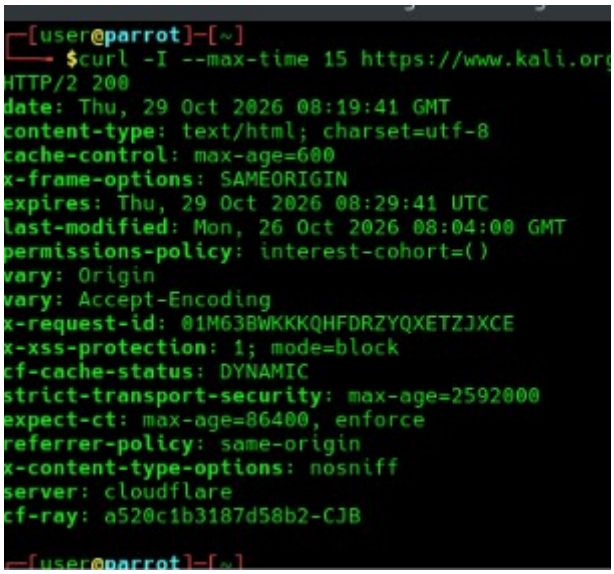
```
getent hosts kali.org
```

Result:



Test 4: HTTPS Connectivity

```
curl -I --max-time 15 https://www.kali.org/
```



Screenshot 9: *HTTPS connectivity test result.*

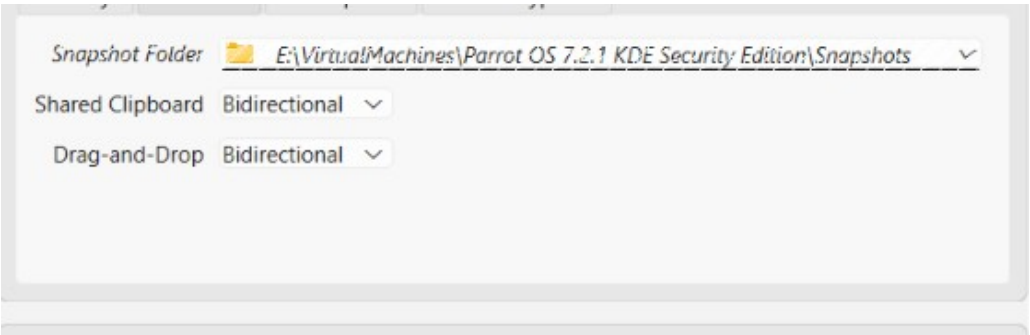
Conclusion: Internet connectivity was [verified / not yet verified] based on the test results recorded above.

8. SHARED CLIPBOARD AND DRAG-AND-DROP CONFIGURATION

Shared clipboard and drag-and-drop were configured to support data exchange between the Windows host and the Kali Linux virtual machine.

Procedure

1. Opened the Kali Linux virtual machine settings in VirtualBox.
2. Navigated to General → Advanced.
3. Set Shared Clipboard to Bidirectional.
4. Set Drag'n'Drop to Bidirectional.
5. Saved the settings.
6. Tested text transfer between the host and guest operating systems.



Screenshot 10: *Shared clipboard and drag-and-drop settings.*

9. SHARED DOWNLOADS FOLDER CONFIGURATION

The Windows Downloads folder was configured as a shared folder for Kali Linux.

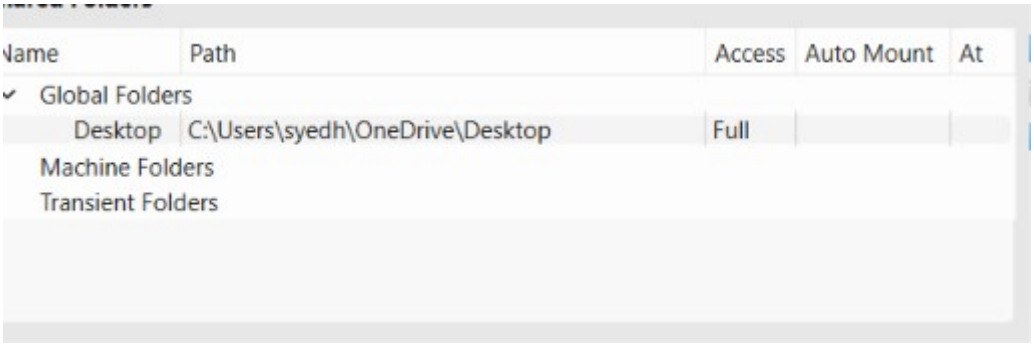
Procedure

1. Opened Kali Linux virtual machine settings.
2. Selected Shared Folders.
3. Added the Windows Downloads folder.
4. Set the shared folder name to downloads.
5. Enabled Auto-mount and Make Permanent.
6. Saved the settings.
7. Checked the shared-folder mount inside Kali Linux.

Verification Command

```
ls -la /media/sf_downloads
```

Observed Result: [Record whether the shared folder was mounted and accessible.]



Screenshot 11: *VirtualBox shared-folder configuration.*

10. VIRTUAL MACHINE SNAPSHOT

A snapshot was created to preserve the configured Kali Linux lab environment.

Procedure

1. Opened VirtualBox Manager.
2. Selected the Kali Linux virtual machine.
3. Opened the Snapshots view.
4. Created a snapshot named WK1-PM1-Clean-Lab-Setup.
5. Added a description of the lab configuration.
6. Verified that the snapshot appeared in the snapshot list.

The snapshot provides a restore point for the configured virtual machine before future cybersecurity exercises.

Screenshot 13: *Insert a screenshot showing the saved Kali Linux snapshot in VirtualBox.*

11. FINAL VERIFICATION CHECKLIST

Requirement	Status
VirtualBox installed	[Completed / Pending]
Kali Linux imported and booted	[Completed / Pending]
Custom NAT Network created	[Completed / Pending]
Network subnet configured as 10.0.0.0/24	[Completed / Pending]
Kali Linux static IP configured as 10.0.0.2/24	[Completed / Pending]
Internet connectivity verified	[Completed / Pending]
Shared clipboard enabled and tested	[Completed / Pending]
Drag-and-drop enabled and tested	[Completed / Pending]
Windows Downloads folder shared and accessible	[Completed / Pending]
Virtual machine snapshot created	[Completed / Pending]

12. CONCLUSION

The WK1-PM1 task focuses on creating a virtualized cybersecurity testing environment using Oracle VirtualBox and Kali Linux.

The required lab configuration includes a custom NAT Network using the 10.0.0.0/24 subnet, the specified static IP address for Kali Linux, internet connectivity, host–guest integration, a shared Downloads folder, and a virtual machine snapshot.

The final status of the lab is based on the verification results and screenshots included in this report. Any requirement that remains incomplete will be identified and resolved before final submission.

END OF REPORT

Prepared by: Syed Raihaan